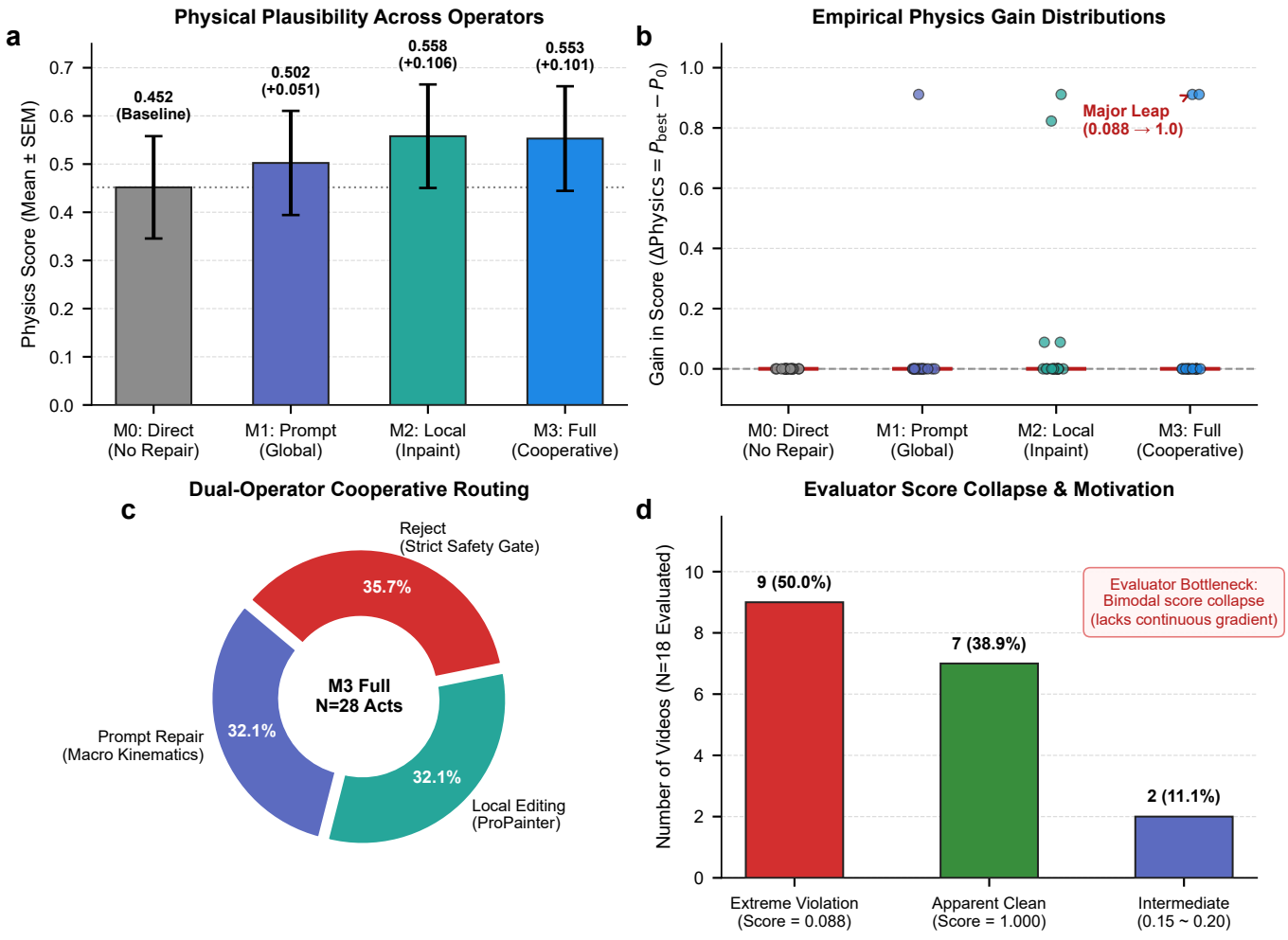

a
Physical Plausibility Across Operators
Physics Score (Mean ± SEM)
0.452
(Baseline)
0.502
(+0.051)
0.558
(+0.106)
0.553
(+0.101)
M0: Direct
(No Repair)
M1: Prompt
(Global)
M2: Local
(Inpaint)
M3: Full
(Cooperative)
b
Empirical Physics Gain Distributions
Gain in Score (ΔPhysics = P_best − P_0)
Major Leap
(0.088 → 1.0)
c
Dual-Operator Cooperative Routing
Reject
(Strict Safety Gate)
35.7%
M3 Full
N=28 Acts
Prompt Repair
(Macro Kinematics)
32.1%
32.1%
Local Editing
(ProPainter)
d
Evaluator Score Collapse & Motivation
Number of Videos (N=18 Evaluated)
9 (50.0%)
7 (38.9%)
2 (11.1%)
Evaluator Bottleneck:
Bimodal score collapse
(lacks continuous gradient)
Extreme Violation
(Score = 0.088)
Apparent Clean
(Score = 1.000)
Intermediate
(0.15 ~ 0.20)